

Supporting Information

Activating the Gate-Opening of a Metal–Organic Framework and Maximizing Its Adsorption Capacity

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Materials and characterization: All chemicals and solvents were purchased from commercial sources and used without modification or purification. Thermogravimetric analysis was acquired using a Shimadzu TGA-50 system under a N₂ atmosphere at a heating rate of 5°C min⁻¹ to determine the amount of H₂O in the metal source. Deionized water was obtained using a Millipore Direct-Q3 UV water purification system. Unless otherwise noted, all samples were pre-dried under vacuum at room temperature for 3 h, followed by dynamic vacuum pretreatment at 250°C for 3 h to ensure identical experimental conditions prior to analysis. Scanning electron microscopy (SEM) images were obtained using a JEOL IT-500HR field-emission SEM and a JEOL JSM-7610F-Plus field-emission SEM. Powder X-ray diffraction (PXRD) patterns were obtained using a Rigaku MiniFlex 600 instrument equipped with Cu K α radiation source (40 kV, 15 mV). MDI JADE 9.0 software (Materials Data Inc., Livermore, CA, USA) was used to calculate the cell parameters. Crystallmaker software (version 2.2.2) was used to render the schematic representation (ball-and-stick version) of the samples. The simulated structures of the samples were obtained from the Cambridge Crystallographic Data Center. For samples without available simulated structures, modifications were made using the calculated cell parameters in the VESTA software package. ¹H NMR spectra were recorded using a Bruker Avance III HD 300 spectrometer (¹H NMR, 300 MHz) with acetic acid-*d*₄ and dimethyl sulfoxide (DMSO)-*d*₆. Infrared (IR) spectra of the solid samples were measured utilizing a Jasco FT/IR 4200 spectrometer equipped with the attenuated total reflection module. The sorption isotherms of N₂ (77 K) were measured in the gaseous state using BELSORP Max and BELSORP II-mini volumetric adsorption equipment. The pore-size distribution was calculated using the non-local density functional theory method with BELMaster version 6.3.0.0 software. The sorption isotherms, heat of adsorption, and rate of adsorption of CO₂ (195 K, 273K, and 298K) were

measured in the gaseous state using BELSORP Max with BELMaster version 7.4.1.1 software. Energy-dispersive X-ray spectroscopy (EDX) spectra were acquired using a Hitachi SU 1510 SEM device equipped with a Horiba EMAX Energy E-250 EDX system. 2-Chloroethyl ethyl sulfide (CEES) adsorption measurements were performed with a custom-made jar-in-jar setup. The kinetic parameters for CEES adsorption were determined using the nonlinear curve fitting program in OriginPro version 8.6 software.

Preparation of 4,4'-biphenyldicarboxylic acid (H₂BPDC) solution: A solution of H₂BPDC was prepared by adding the required amount of H₂BPDC powder to *N,N*-dimethylformamide (DMF) to achieve the desired concentration. The mixture was then sonicated to form a homogeneous mixture then heated in an oil bath at 120°C for 30 min. When the solution became transparent, it was cooled to room temperature for further use.

Synthesis of In-MIL-53D: A 0.055 M solution of In(NO₃)₃·*x*H₂O (*x* = 3, determined by TGA; 156 mg, 0.44 mmol) and a 0.045 M solution of H₂BPDC (87 mg, 0.36 mmol) were mixed in DMF (16 mL). The mixture was transferred to a 70 mL vial, placed in an oil bath, and heated at 120°C for 30 min. The product was then cooled to room temperature, isolated, and washed with fresh DMF and methanol using a centrifugation–redispersion cycle.

Synthesis of In-MIL-53D hybrids, Hybrid-X (X = 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, and 5):

In(NO₃)₃·*x*H₂O (156 mg, 0.44 mmol), H₂BPDC (0.36 × (1–*x*) mmol), and 1,4-benzenedicarboxylic acid (H₂BDC; 0.36 × *x* mmol) were dissolved in DMF (16 mL) (where *x* = 0.02, 0.07, 0.12, 0.17, 0.22, 0.27, 0.32, 0.37, 0.42, and 0.47). The mixture was transferred to a 70 mL vial, placed in an oil bath, and heated at 120°C for 30 min. The product was cooled to room

temperature, isolated, and washed with fresh DMF and methanol using a centrifugation–redispersion cycle.

Synthesis of In-MIL-68B: A 0.055 M solution of $\text{In}(\text{NO}_3)_3 \cdot x\text{H}_2\text{O}$ (156 mg, 0.44 mmol) and a 0.045 M solution of H_2BDC (60 mg, 0.36 mmol) were mixed in DMF (16 mL). The mixture was transferred to a 70 mL vial, placed in an oil bath, and heated at 120°C for 30 min. The product was cooled to room temperature, isolated, and washed with fresh DMF and methanol using a centrifugation–redispersion cycle.

Preparation of 2,6-naphthalenedicarboxylic acid (H_2NDC) solution: A solution of H_2NDC was prepared by adding the required amount of H_2NDC powder to DMF to achieve the desired concentration. The mixture was then sonicated to form a homogeneous mixture then heated in an oil bath at 120°C for 5 min. When the solution became transparent, it was cooled to room temperature for further use.

Synthesis of Fe-MIL-88C: A 0.055 M solution of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (178 mg, 0.44 mmol) and a 0.045 M solution of H_2NDC (78 mg, 0.36 mmol) were mixed in DMF (16 mL). The mixture was transferred to a 70 mL vial, placed in an oil bath, and heated at 120°C for 30 min. The product was then cooled to room temperature, isolated, and washed with fresh DMF and methanol using a centrifugation–redispersion cycle.

Synthesis of Fe-MIL-88C hybrid (X%) ($X = 12\%$, and 26%): $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (178 mg, 0.44 mmol), H_2NDC ($0.36 \times (1-x)$ mmol), and H_2BDC ($0.36 \times x$ mmol) were dissolved in DMF (16 mL) (where $x = 0.17$ and 0.37). The mixture was transferred to a 70 mL vial, placed in an oil

bath, and heated at 120°C for 30 min. The product was cooled to room temperature, isolated, and washed with fresh DMF and methanol using a centrifugation–redispersion cycle.

Adsorption of CEES: The adsorption of CEES was conducted at 23°C using a custom-made jar-in-jar setup. For the experiments, In-MIL-53D and hybrid In-MIL-53D powders (2.0 mg) were placed in a ceramic crucible pan, while CEES (2.0 mL) was placed in a 5 mL beaker. The crucible pan and beaker were positioned inside a large jar, which was sealed with a lid. After exposure to CEES vapor for defined time intervals, the samples were dissolved in acetic acid- d_4 and DMSO- d_6 and analyzed by ^1H NMR spectroscopy to quantify the amount of CEES adsorbed by each sample.

Adsorption kinetics of Hybrid-X: The adsorption of CEES was fitted to a pseudo-first-order model using the equation:

$$q_t = q_e(1 - e^{-kt}),$$

where, q_e is equilibrium adsorption capacity (mg g^{-1}), q_t is the adsorption capacity (mg g^{-1}) at time t , and k is the pseudo-first-order rate constant (h^{-1}).

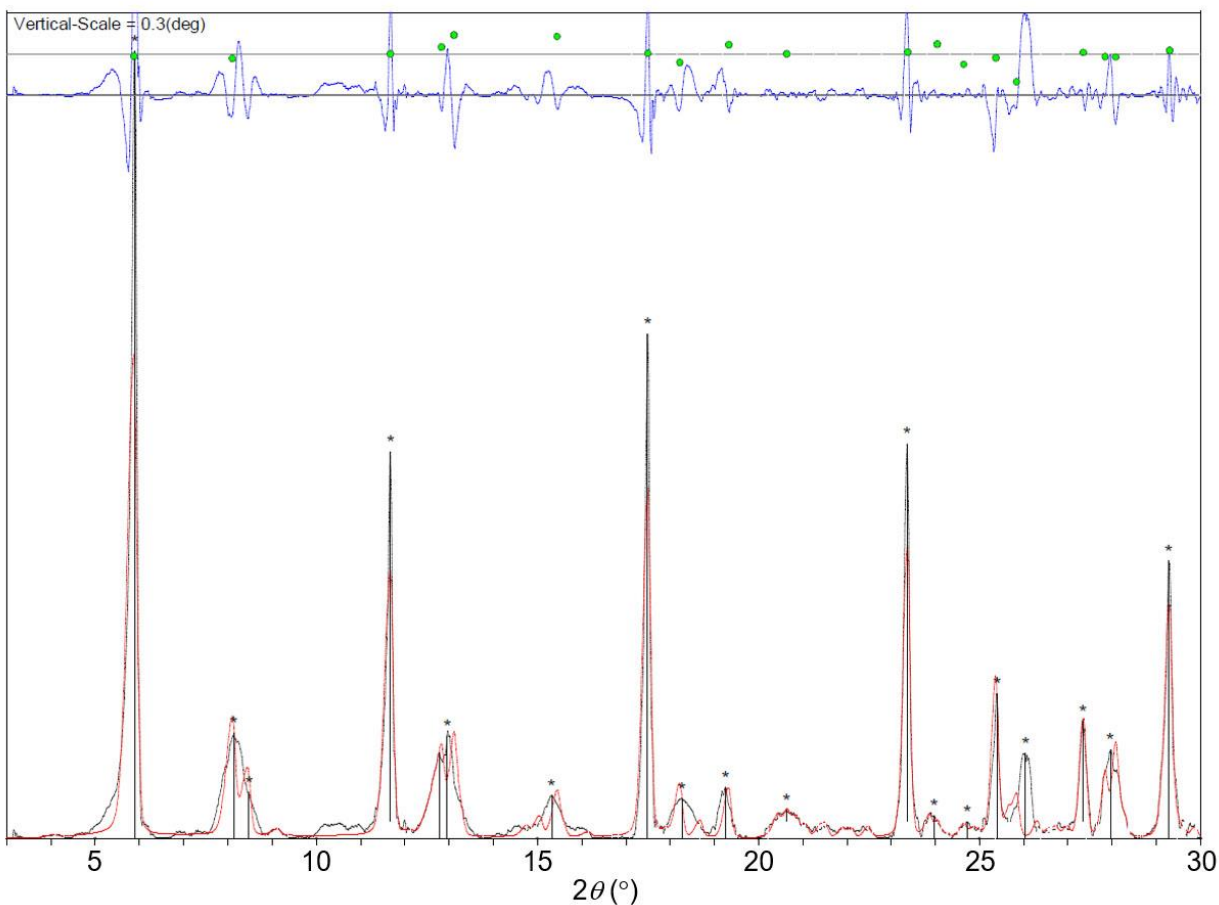


Figure S1. Whole pattern fitting of In-MIL-53D soaked in DMF (never dried). The black line is the experimental PXRD pattern and the red line is the calculated PXRD pattern. The top blue line indicates the deviation between the experimental and calculated PXRD patterns. The unit cell parameters of In-MIL-53D soaked in DMF (never dried) calculated using MDI Jade 9.0 software were $a = 22.07 \text{ \AA}$, $b = 21.17 \text{ \AA}$, $c = 7.14 \text{ \AA}$, and $\alpha = \beta = \gamma = 90^\circ$.

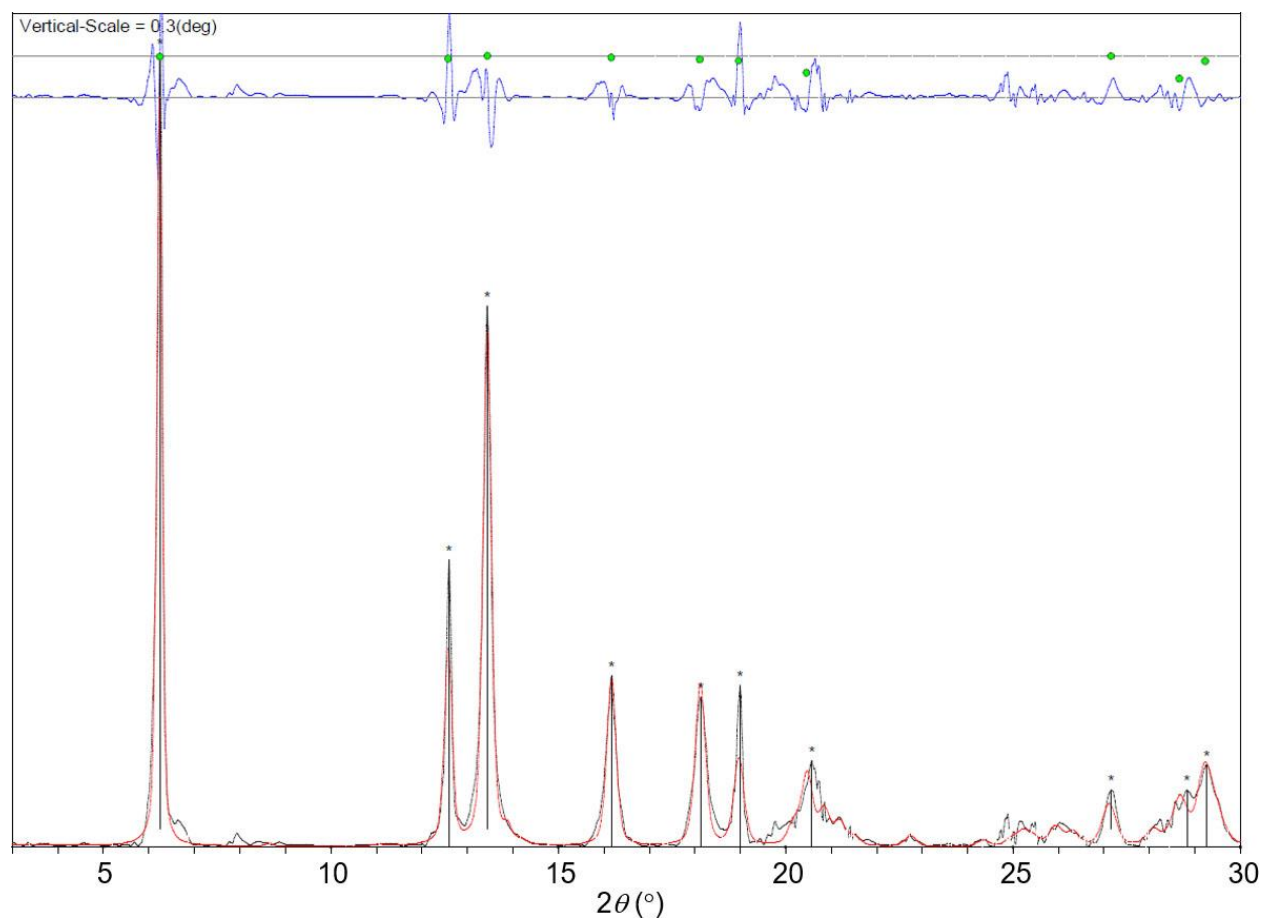


Figure S2. Whole pattern fitting of In-MIL-53D dried at 250°C for 3 h. The black line is the experimental PXRD pattern and the red line is the calculated PXRD pattern. The top blue line indicates the deviation between the experimental and calculated PXRD patterns. The unit cell parameters of In-MIL-53D dried at dried at 250°C for 3 h calculated using MDI Jade 9.0 software were $a = 25.46 \text{ \AA}$, $b = 16.70 \text{ \AA}$, $c = 7.11 \text{ \AA}$, and $\alpha = \beta = \gamma = 90^\circ$.

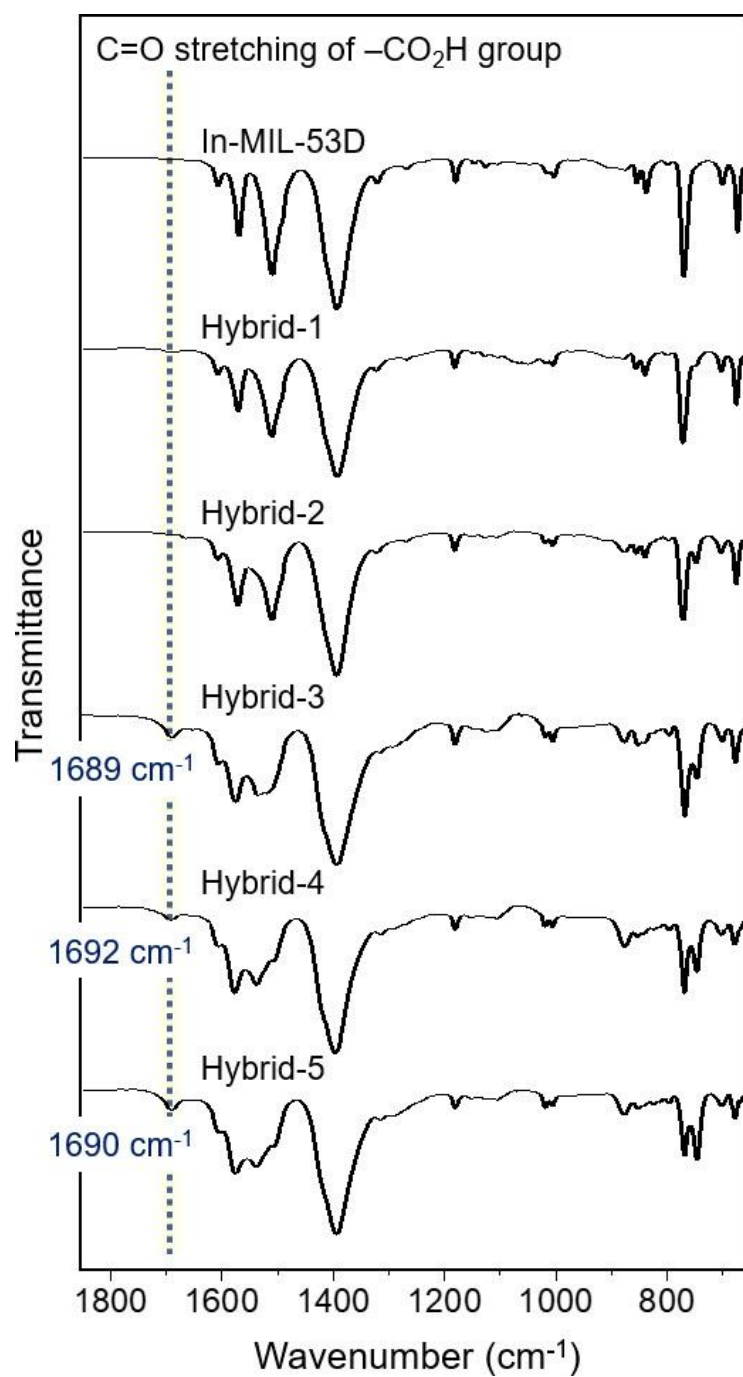


Figure S3. IR spectra of In-MIL-53D, Hybrid-1, Hybrid-2, Hybrid-3, Hybrid-4, and Hybrid-5.

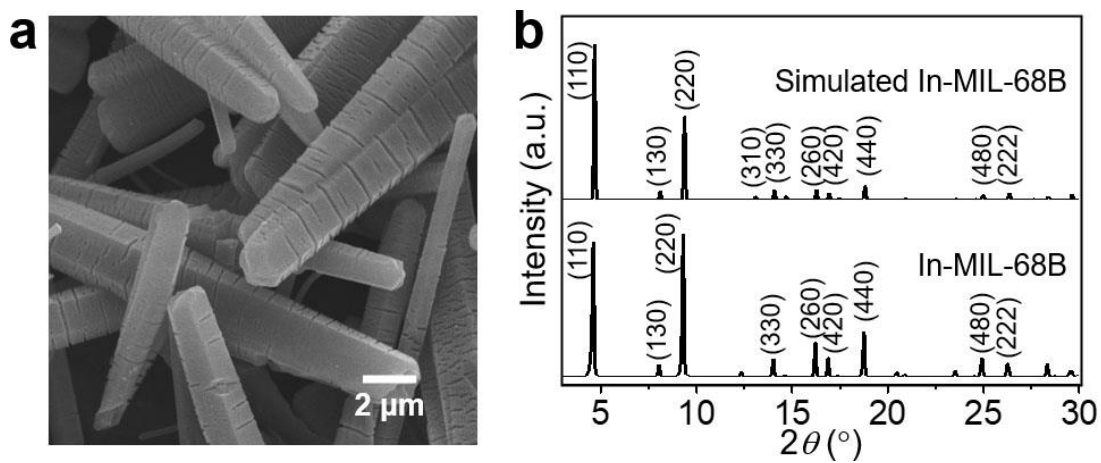


Figure S4. (a) SEM image of In-MIL-68B dried at 250°C for 3 h. (b) PXRD patterns of simulated In-MIL-68B and In-MIL-68B dried at 250°C for 3 h.

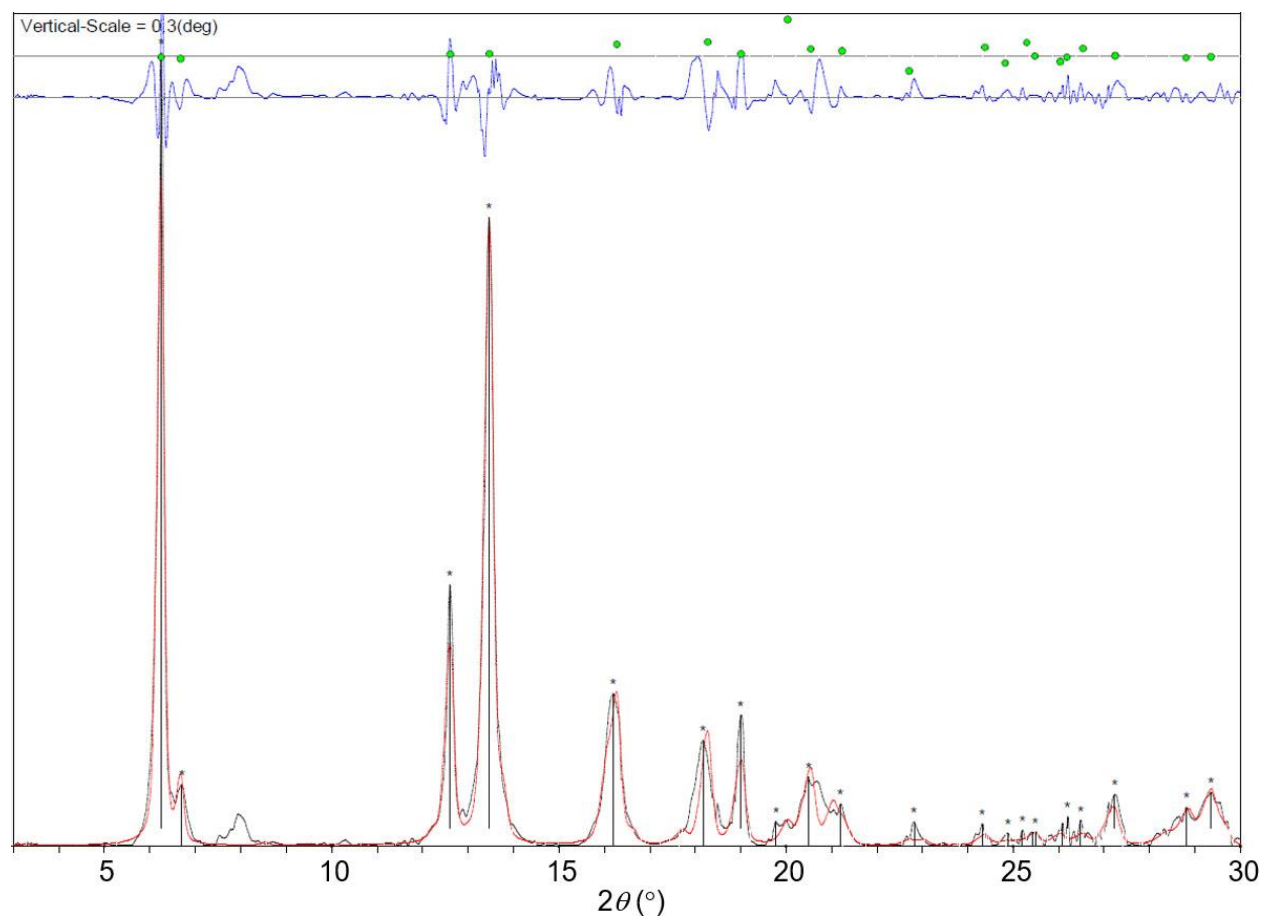


Figure S5. Whole pattern fitting of Hybrid-1 dried at 250°C for 3 h. The black line is the experimental PXRD pattern and the red line is the calculated PXRD pattern. The top blue line indicates the deviation between the experimental and calculated PXRD patterns. The unit cell parameters of Hybrid-1 dried at 250°C for 3 h calculated using MDI Jade 9.0 software were $a = 26.08 \text{ \AA}$, $b = 16.46 \text{ \AA}$, $c = 7.17 \text{ \AA}$, and $\alpha = \beta = \gamma = 90^{\circ}$.

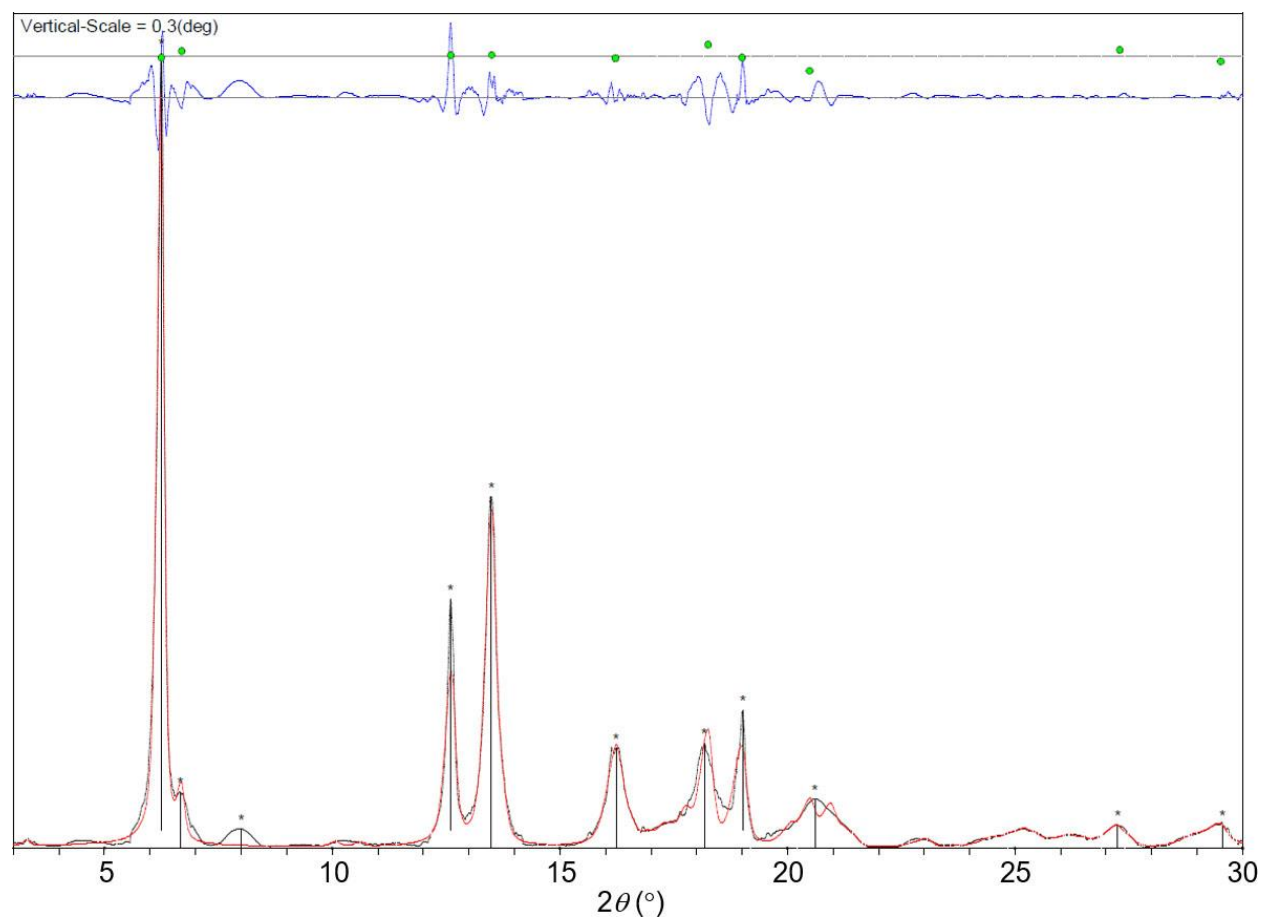


Figure S6. Whole pattern fitting of Hybrid-2 dried at 250°C for 3 h. The black line is the experimental PXRD pattern and the red line is the calculated PXRD pattern. The top blue line indicates the deviation between the experimental and calculated PXRD patterns. The unit cell parameters of Hybrid-2 dried at 250°C for 3 h calculated using MDI Jade 9.0 software were $a = 26.02 \text{ \AA}$, $b = 16.49 \text{ \AA}$, $c = 7.21 \text{ \AA}$, and $\alpha = \beta = \gamma = 90^\circ$.

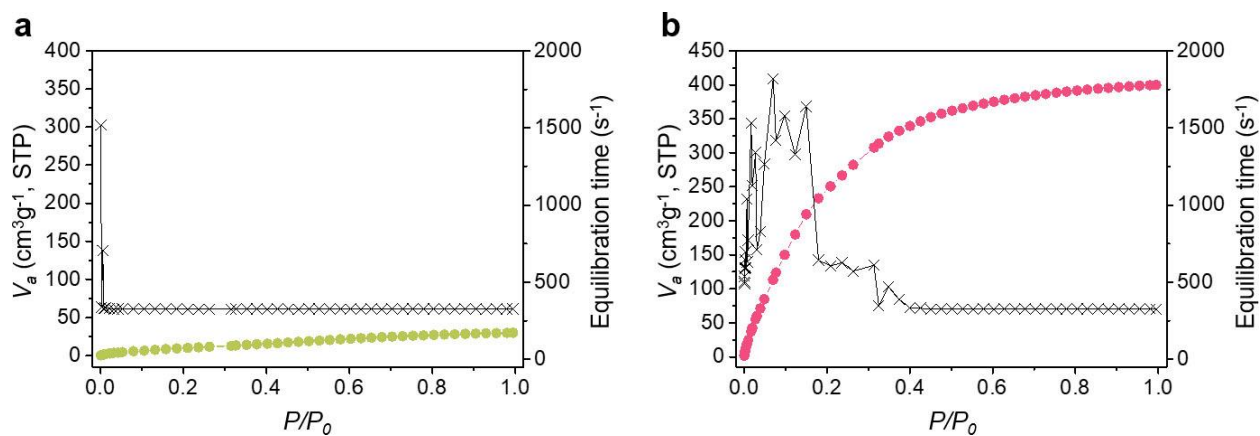


Figure S7. Adsorption of CO₂ on (a) In-MIL-53D and (b) Hybrid-4 at 195 K. Filled symbols correspond to the sorption isotherms, while the cross symbols correspond to the equilibration time for each data point of the sorption isotherm.

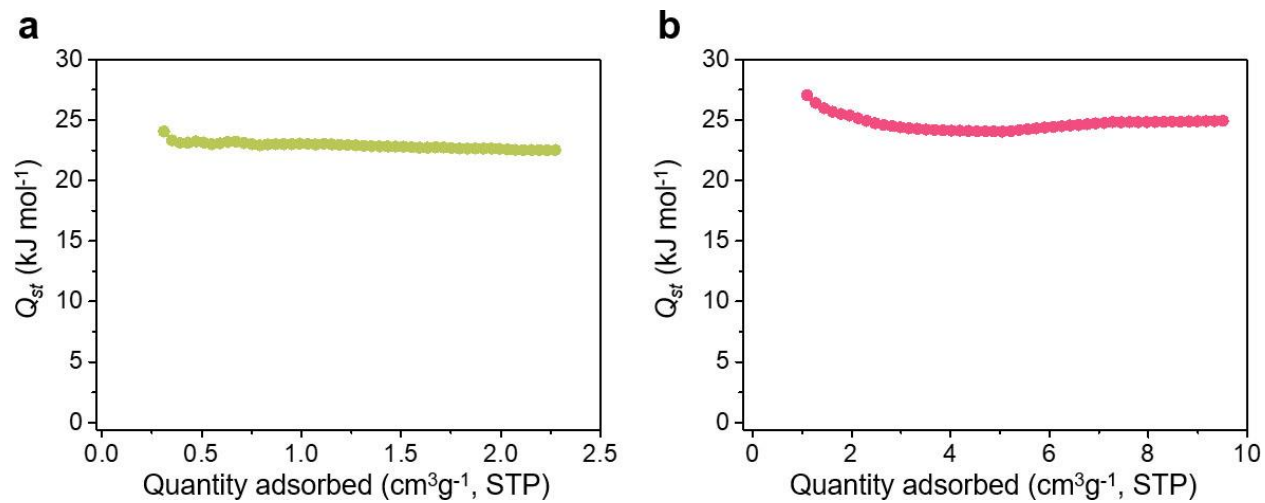


Figure S8. The isosteric heat of adsorption (Q_{st}) for CO_2 of (a) In-MIL-53D and (b) Hybrid-4.

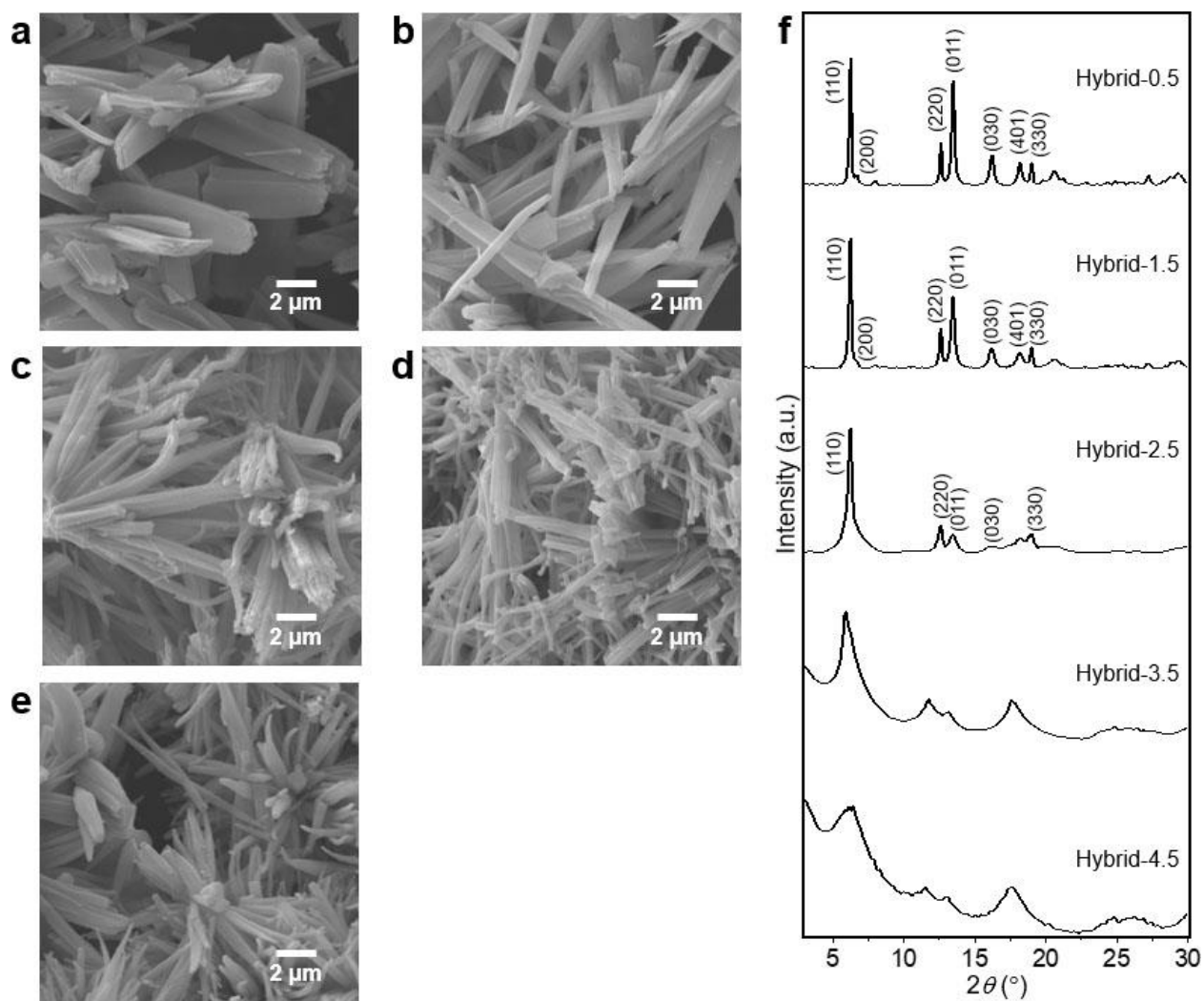


Figure S9. SEM images of (a) Hybrid-0.5, (b) Hybrid-1.5, (c) Hybrid-2.5, (d) Hybrid-3.5, and (e) Hybrid-4.5. (f) PXRD patterns of a series of defect-containing In-MIL-53D hybrids.

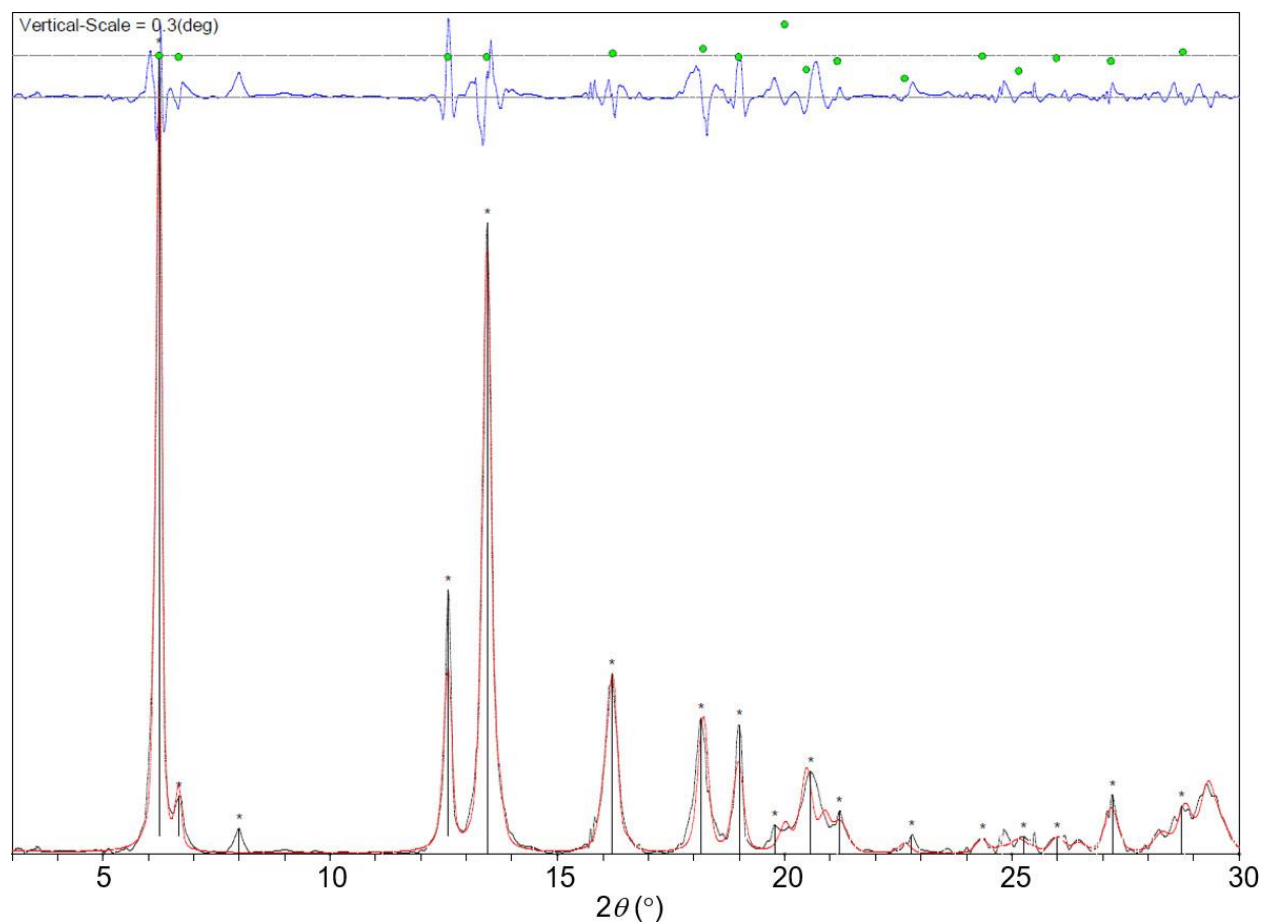


Figure S10. Whole pattern fitting of Hybrid-0.5 dried at 250°C for 3 h. The black line is the experimental PXRD pattern and the red line is the calculated PXRD pattern. The top blue line indicates the deviation between the experimental and calculated PXRD patterns. The unit cell parameters of Hybrid-0.5 dried at 250°C for 3 h calculated using MDI Jade 9.0 software were $a = 26.14 \text{ \AA}$, $b = 16.50 \text{ \AA}$, $c = 7.22 \text{ \AA}$, and $\alpha = \beta = \gamma = 90^\circ$.

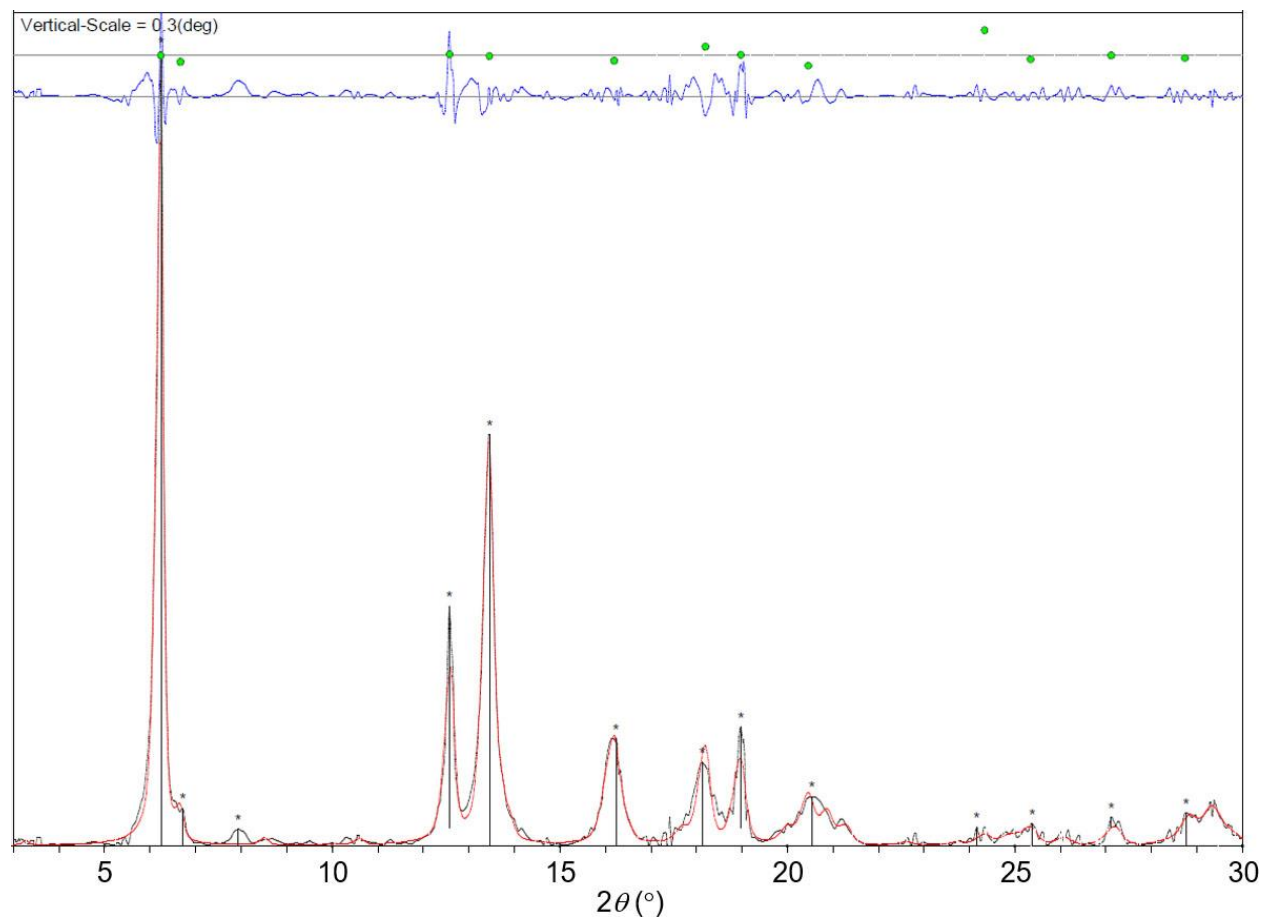


Figure S11. Whole pattern fitting of Hybrid-1.5 dried at 250°C for 3 h. The black line is the experimental PXRD pattern and the red line is the calculated PXRD pattern. The top blue line indicates the deviation between the experimental and calculated PXRD patterns. The unit cell parameters of Hybrid-1.5 dried at 250°C for 3 h calculated using MDI Jade 9.0 software were $a = 26.12 \text{ \AA}$, $b = 16.50 \text{ \AA}$, $c = 7.23 \text{ \AA}$, and $\alpha = \beta = \gamma = 90^\circ$.

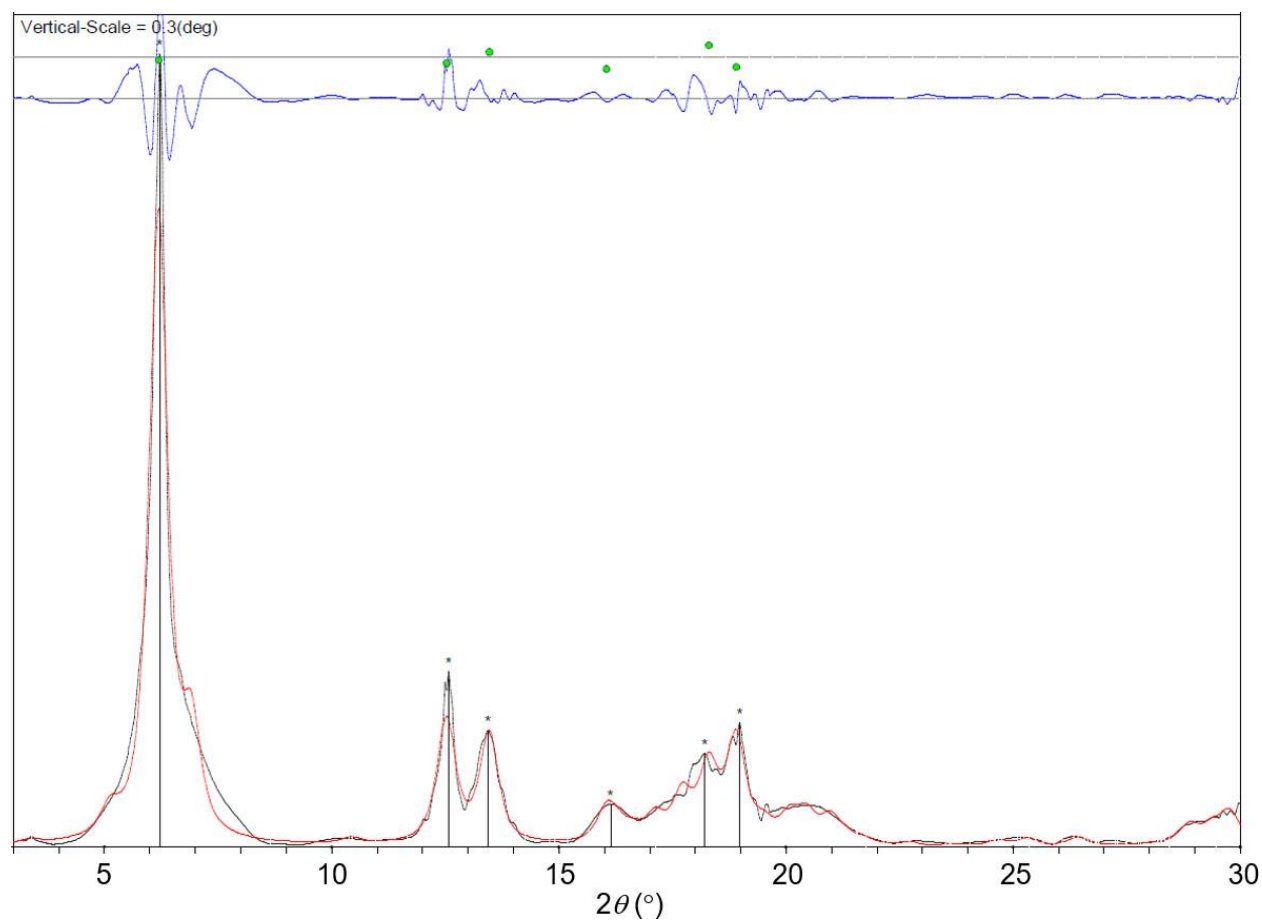


Figure S12. Whole pattern fitting of Hybrid-2.5 dried at 250°C for 3 h. The black line is the experimental PXRD pattern and the red line is the calculated PXRD pattern. The top blue line indicates the deviation between the experimental and calculated PXRD patterns. The unit cell parameters of Hybrid-2.5 dried at 250°C for 3 h calculated using MDI Jade 9.0 software were $a = 25.11 \text{ \AA}$, $b = 16.84 \text{ \AA}$, $c = 7.06 \text{ \AA}$, and $\alpha = \beta = \gamma = 90^\circ$.

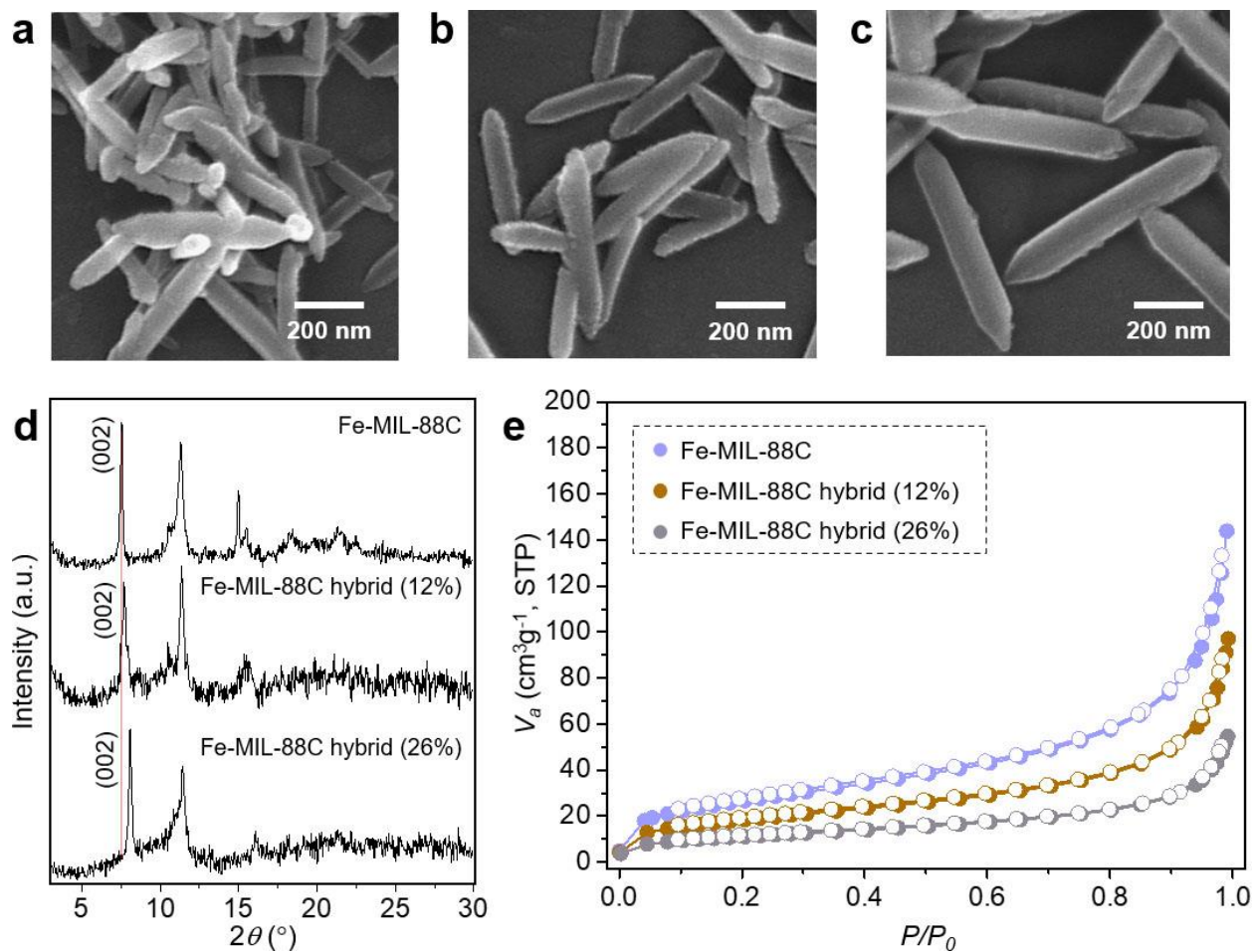


Figure S13. SEM images of (a) Fe-MIL-88C, (b) Fe-MIL-88C hybrid (12%), and (c) Fe-MIL-88C hybrid (26%). (d) PXRD patterns and (e) N₂ sorption isotherms of Fe-MIL-88C and Fe-MIL-88C hybrids.

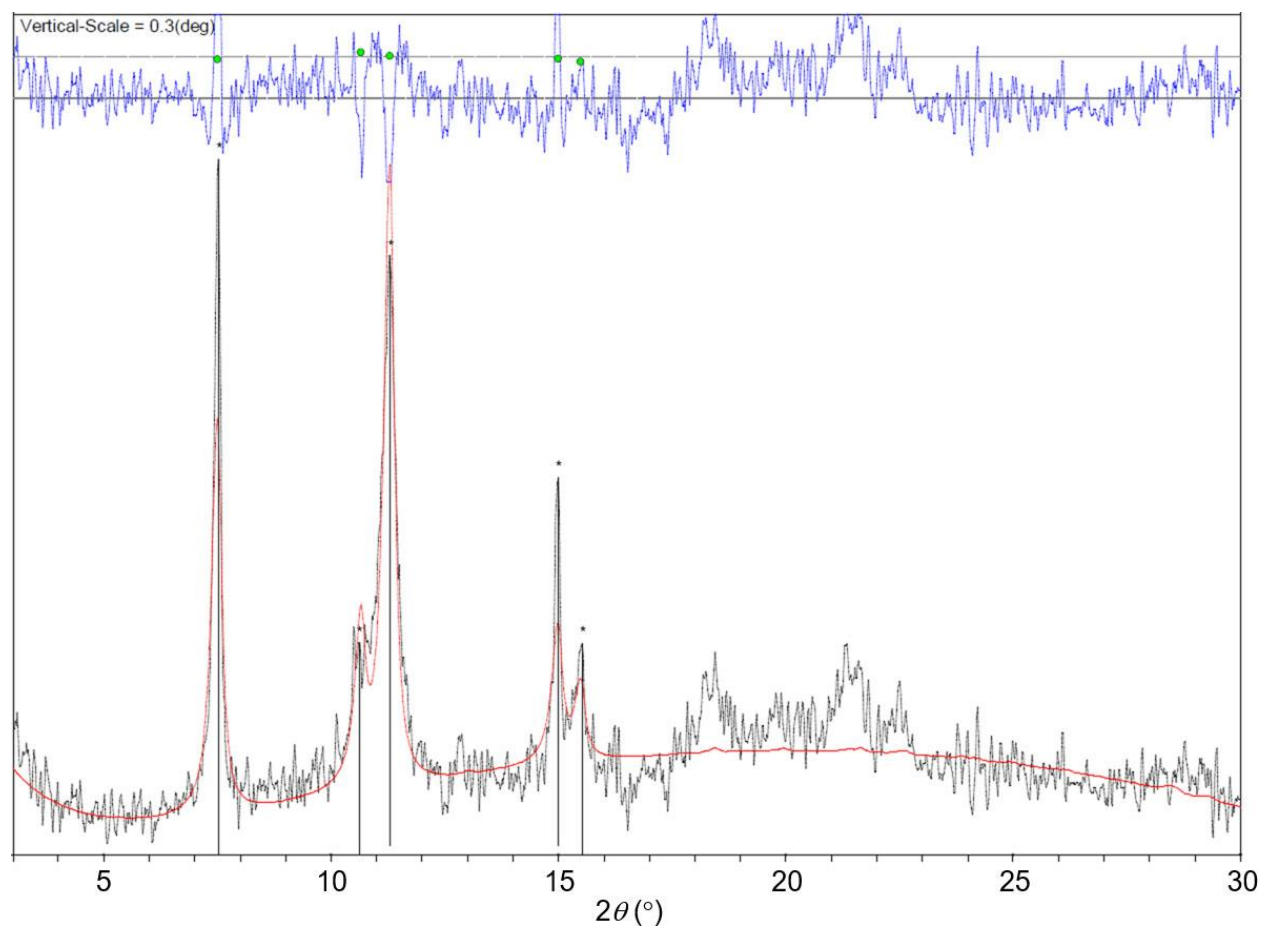


Figure S14. Whole pattern fitting of Fe-MIL-88C dried at 150°C for 3 h. The black line is the experimental PXRD pattern and the red line is the calculated PXRD pattern. The top blue line indicates the deviation between the experimental and calculated PXRD patterns. The unit cell parameters of Fe-MIL-88C dried at 150°C for 3 h calculated using MDI Jade 9.0 software were $a = b = 9.63 \text{ \AA}$, $c = 23.71 \text{ \AA}$, $\alpha = \beta = 90^\circ$, and $\gamma = 120^\circ$.

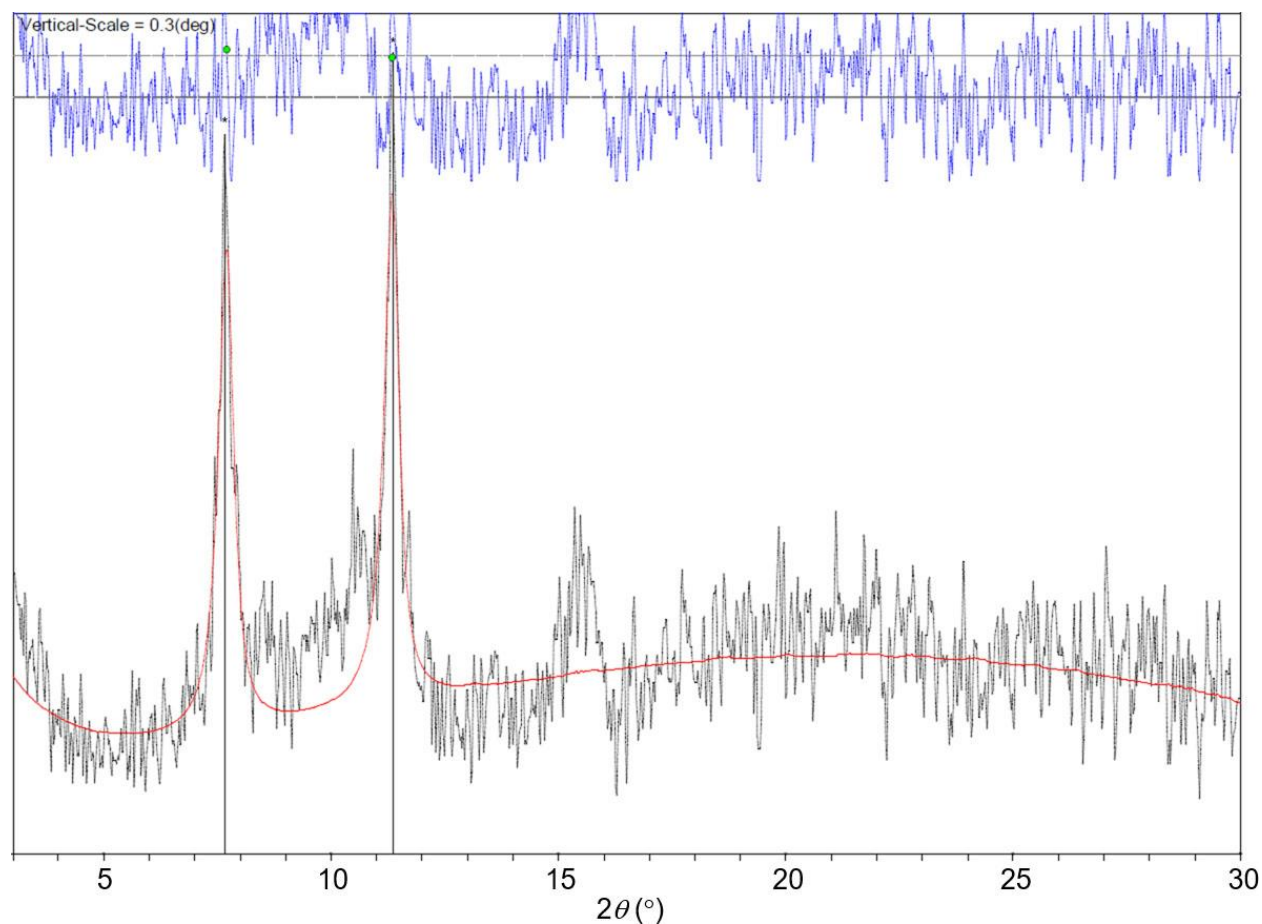


Figure S15. Whole pattern fitting of Fe-MIL-88C hybrid (12%) dried at 150°C for 3 h. The black line is the experimental PXRD pattern and the red line is the calculated PXRD pattern. The top blue line indicates the deviation between the experimental and calculated PXRD patterns. The unit cell parameters of Fe-MIL-88C hybrid (12%) dried at 150°C for 3 h calculated using MDI Jade 9.0 software were $a = b = 9.63 \text{ \AA}$, $c = 23.17 \text{ \AA}$, $\alpha = \beta = 90^\circ$, and $\gamma = 120^\circ$.

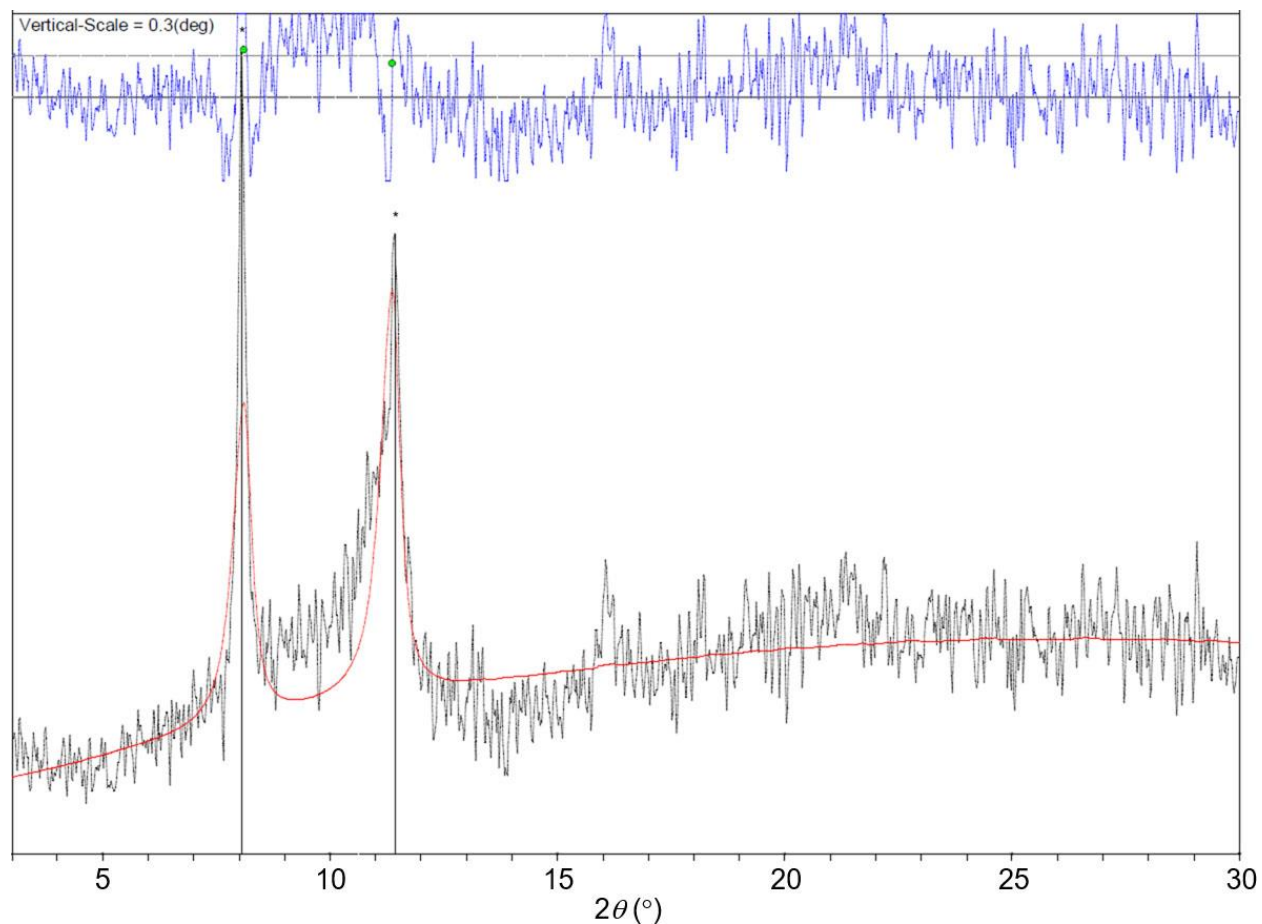


Figure S16. Whole pattern fitting of Fe-MIL-88C hybrid (26%) dried at 150°C for 3 h. The black line is the experimental PXRD pattern and the red line is the calculated PXRD pattern. The top blue line indicates the deviation between the experimental and calculated PXRD patterns. The unit cell parameters of Fe-MIL-88C hybrid (26%) dried at 150°C for 3 h calculated using MDI Jade 9.0 software were $a = b = 9.66 \text{ \AA}$, $c = 21.98 \text{ \AA}$, $\alpha = \beta = 90^\circ$, and $\gamma = 120^\circ$.

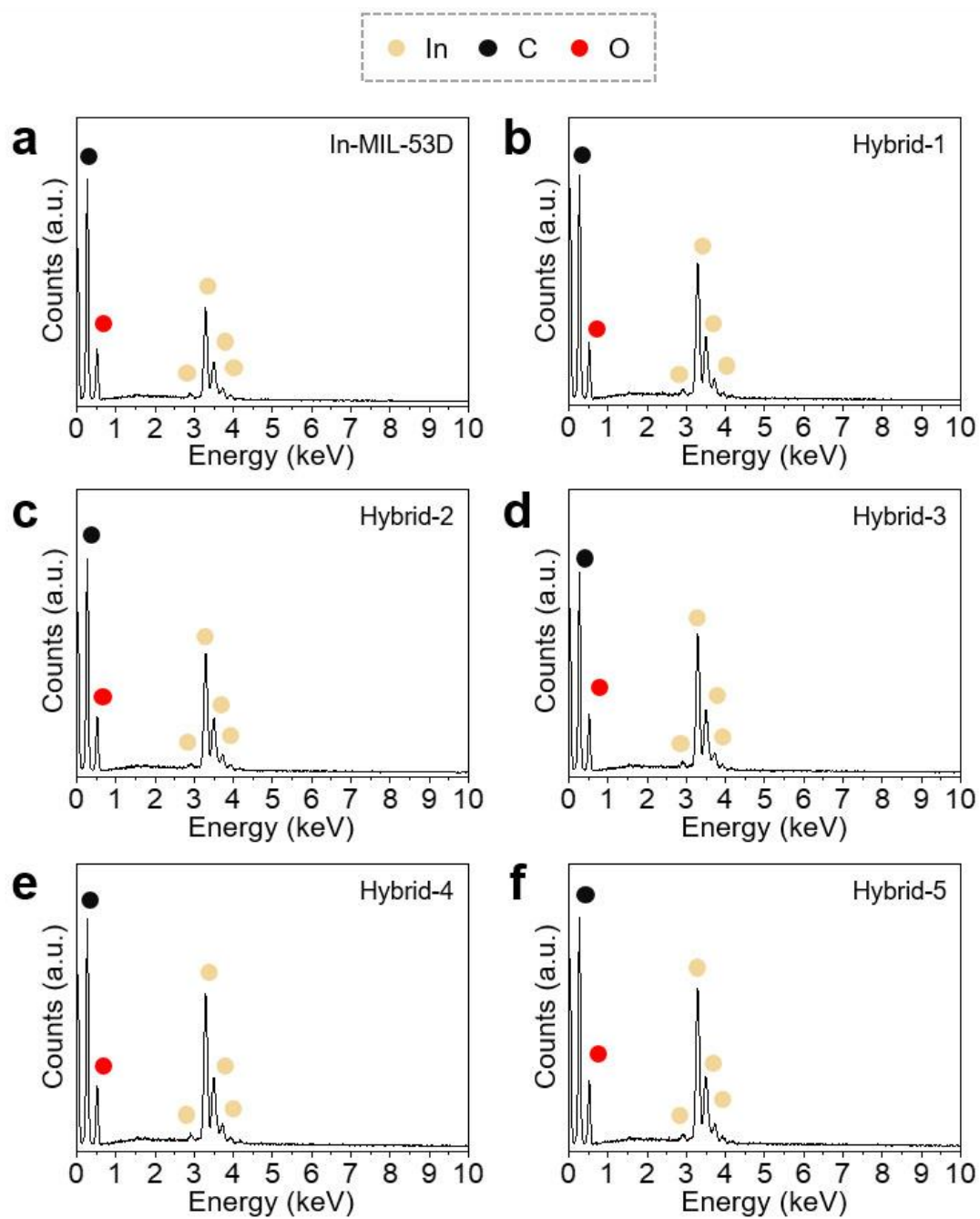


Figure S17. EDX spectra of (a) In-MIL-53D, (b) Hybrid-1, (c) Hybrid-2, (d) Hybrid-3, (e) Hybrid-4, and (f) Hybrid-5.

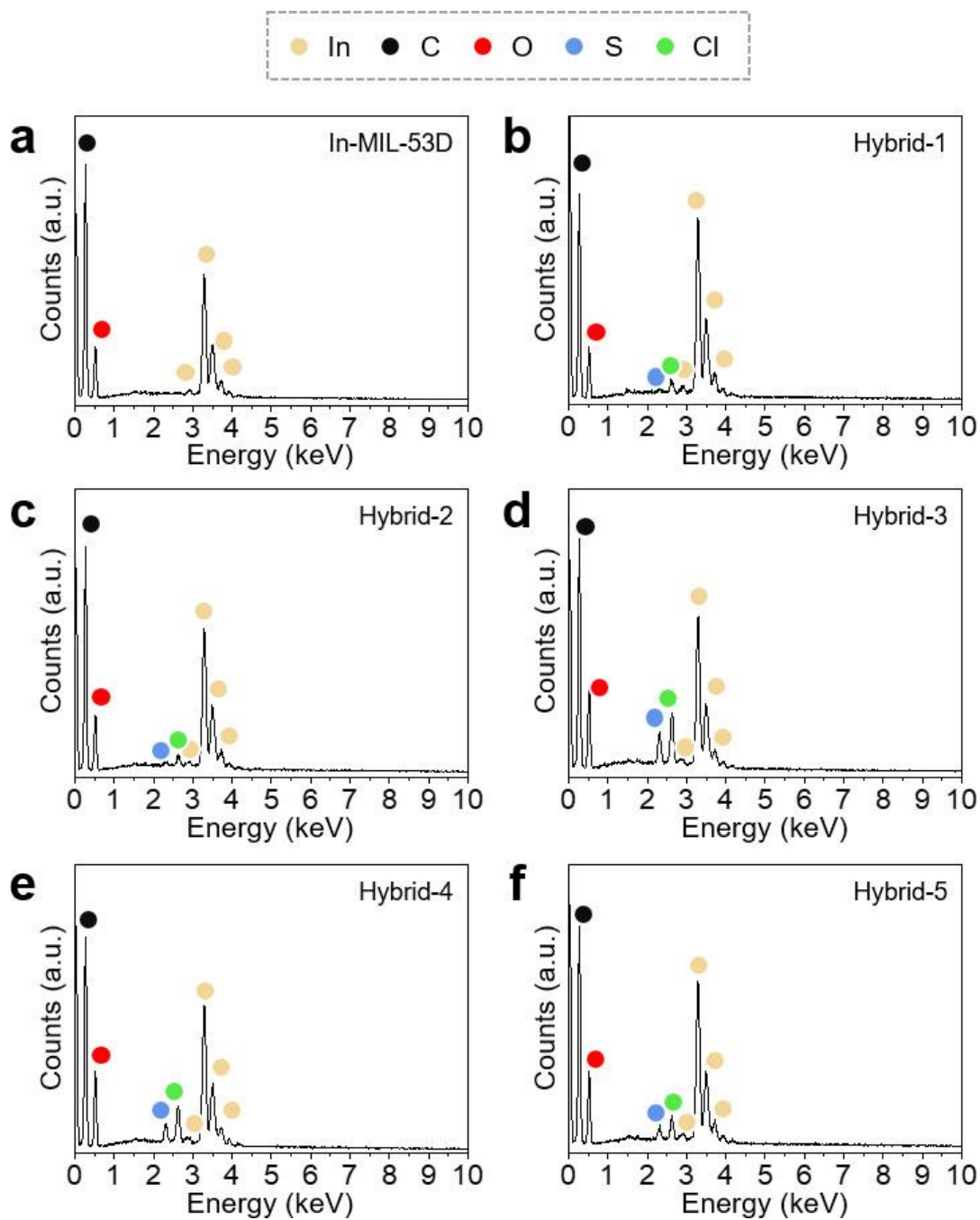


Figure S18. EDX spectra of (a) In-MIL-53D, (b) Hybrid-1, (c) Hybrid-2, (d) Hybrid-3, (e) Hybrid-4, and (f) Hybrid-5 measured after exposure to CEES vapors for 240 min.

Table S1. Ratios of the H₂BPDC and H₂BDC linkers used in the solvothermal reaction of In-MIL-53D and the series of defect-containing In-MIL-53D hybrids and ratios of the incorporated BPDC²⁻ and HBDC⁻, as determined by analyzing the ¹H NMR spectra of the samples.

MOF	Used ratio (%)		Averaged ^a incorporated ratio (%)	
	H ₂ BPDC	H ₂ BDC	BPDC ²⁻	HBDC ⁻
In-MIL-53D	100	0	100	0
Hybrid-0.5	98	2	95	5
Hybrid-1	93	7	90	10
Hybrid-1.5	88	12	85	15
Hybrid-2	83	17	80	20
Hybrid-2.5	78	22	75	25
Hybrid-3	73	27	70	30
Hybrid-3.5	68	32	64	36
Hybrid-4	63	37	61	39
Hybrid-4.5	58	42	55	45
Hybrid-5	53	47	50	50

^aAveraged incorporated ratio (%) is based on four independent experiments.

Table S2. Maximum N₂ uptake capacities, Brunauer–Emmett–Teller (BET) surface areas, and total pore volumes of In-MIL-53D and the series of defect-containing In-MIL-53D hybrids.

MOF	Uptake ($P/P_0 = 1.0$) [cm ³ g ⁻¹]	$a_{s,BET}$ [m ² g ⁻¹]	Total pore volume ($P/P_0 = 0.990$) [cm ³ g ⁻¹]
In-MIL-53D	62.6	77.7	0.097
Hybrid-0.5	59.4	124.4	0.091
Hybrid-1	88.4	196.2	0.134
Hybrid-1.5	155.1	467.4	0.238
Hybrid-2	192.0	649.1	0.295
Hybrid-2.5	280.6	776.2	0.433
Hybrid-3	329.2	1033.7	0.509
Hybrid-3.5	427.1	1373.9	0.657
Hybrid-4	484.1	1680.2	0.747
Hybrid-4.5	483.4	1651.2	0.746
Hybrid-5	459.8	1534.8	0.710

Table S3. Maximum CO₂ uptake capacities of In-MIL-53D and the series of defect-containing In-MIL-53D hybrids at 1 atm and 195 K.

MOF	Uptake ($P/P_0 = 1.0$) [cm ³ g ⁻¹]
In-MIL-53D	32.1
Hybrid-1	58.8
Hybrid-2	134.3
Hybrid-3	220.9
Hybrid-4	351.5
Hybrid-5	302.6

Table S4. Comparison of maximum CO₂ uptake capacities of MOFs at 1 atm and 195 K.

MOF	Uptake ($P/P_0 = 1.0$) [cm ³ g ⁻¹]	Reference
Fe ₂ (BPEB) ₃	300	1
SNU-M11	124	2
CPM-107	206	3
Qc-5-Cu-sql-β	59	4
DZU-211	222	5
JUK-8	144	6
NTU-67	91	7
1 _{apohost}	157	8
Fe ₂ (BDP) ₃	298	9
Fe-PF1	343	10
X-dia-1-Ni	325	11
sql-1-Co-NCS	136	12
Zn-DPNA	64	13
DUT-8(Ni)	590	14
CAU-10-CH ₃	88	15
Zn(dcpa)	241	16
CoPzNi	140	17
Co-MOF-β	173	18
1-O ₂	75	19
MOF 1	156	20
JUK-13-SO ₃ H	181	21
MOF-NDC	203	22
mot-5,6-Zn-β	145	23
Zn(hia)(bphy)	82	24
FJI-H14	279	25
JLU-Liu3	337	26
X-pcu-7-Zn	267	27
Cd-MOF	130	28
MIL-68	333	29
Hybrid-4	352	This work

Table S5. Unit cell parameters of Fe-MIL-88C and Fe-MIL-88C hybrids. Unit cell parameters were calculated by whole pattern fitting using MDI Jade 9.0 software. All samples were dried at 150°C for 3 h under vacuum before analyzing.

MOF	Cell parameters	
	$a = b$ (Å)	c (Å)
Fe-MIL-88C	9.63	23.71
Fe-MIL-88C hybrid (12%)	9.63	23.17
Fe-MIL-88C hybrid (26%)	9.66	21.98

Table S6. The average CEES adsorption capacity, expressed as the number of moles at different time points, based on three independent experiments.

MOF	mol _{CEES} /mol _{MOF}					
	15 min	30 min	60 min	120 min	180 min	240 min
In-MIL-53D	0.03	0.03	0.04	0.04	0.05	0.05
Hybrid-1	0.04	0.05	0.07	0.08	0.08	0.09
Hybrid-2	0.05	0.05	0.09	0.12	0.14	0.14
Hybrid-3	0.07	0.09	0.12	0.14	0.15	0.15
Hybrid-4	0.10	0.14	0.19	0.23	0.24	0.23
Hybrid-5	0.02	0.06	0.07	0.09	0.09	0.10

Table S7. Kinetic parameters of the pseudo-first-order model for the adsorption of CEES on In-MIL-53D and the series of defect-containing In-MIL-53D hybrids.

MOF	q_e (mol/mol)	k (h ⁻¹)	R ²
In-MIL-53D	0.05	0.06	0.95
Hybrid-1	0.08	0.04	0.99
Hybrid-2	0.14	0.02	0.99
Hybrid-3	0.15	0.03	0.97
Hybrid-4	0.23	0.03	0.99
Hybrid-5	0.10	0.02	0.99

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